



PSO201/PHY204: Quiz 2

Date: April 3, 2025; Maximum Marks: 20

1. a) Find out the matrix representation of Spin operator $\hat{S}_{\hat{n}} = \frac{\hbar}{2} \hat{n} \cdot \vec{\sigma}$ along an arbitrary direction $\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ where $\vec{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$ are the Pauli matrices. [2]
- b) Consider a spin half particle in the initial state $|\uparrow\rangle$ at $t = 0$. A magnetic field B is applied in the arbitrary direction $\hat{n}$. Use the formula $\exp(iA\hat{n} \cdot \vec{\sigma}) = \cos A \mathbb{I} + i(\hat{n} \cdot \vec{\sigma}) \sin A$ to **directly** calculate the probabilities $P_{\downarrow}$ and $P_{\uparrow}$ of finding the particle to be in $|\downarrow\rangle$ and $|\uparrow\rangle$ states, respectively, at a later time t . [6]
- c) Plot the probabilities $P_{\downarrow}$ and $P_{\uparrow}$ as a function of ωt (where $\omega = \frac{eB}{m}$) for $\theta = \frac{\pi}{4}$. Indicate the maximum and minimum numerical values of the probabilities. [2]

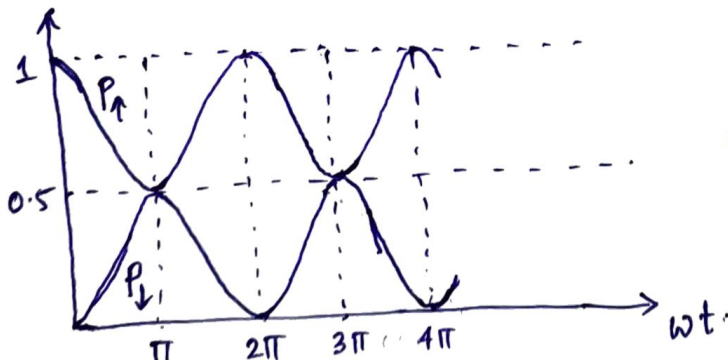
a) $\hat{S}_{\hat{n}} = \frac{\hbar}{2} \left[n_x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + n_y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + n_z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] = \frac{\hbar}{2} \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}$

b) $|\psi(t)\rangle = e^{i\omega \hat{S}_{\hat{n}} t / \hbar} |\uparrow\rangle \quad (\hat{H} = -\omega \hat{S}_{\hat{n}})$
 $\omega = \frac{eB}{m}$
 $= e^{i\frac{\omega t}{\hbar} \frac{\hbar}{2} \hat{n} \cdot \vec{\sigma}} |\uparrow\rangle$
 $= e^{i\frac{\omega t}{2} \hat{n} \cdot \vec{\sigma}} |\uparrow\rangle = \left(\cos \frac{\omega t}{2} \mathbb{I} + i(\hat{n} \cdot \vec{\sigma}) \sin \frac{\omega t}{2} \right) |\uparrow\rangle$
 $= \cos \frac{\omega t}{2} |\uparrow\rangle + i \sin \frac{\omega t}{2} \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
 $= \cos \frac{\omega t}{2} |\uparrow\rangle + i \sin \frac{\omega t}{2} \begin{pmatrix} \cos \theta \\ \sin \theta e^{2i\phi} \end{pmatrix}$
 $= \cos \frac{\omega t}{2} |\uparrow\rangle + 2 \sin \frac{\omega t}{2} \cos \theta |\uparrow\rangle + i \sin \frac{\omega t}{2} \sin \theta e^{2i\phi} |\downarrow\rangle$
 $= \left(\cos \frac{\omega t}{2} + i \sin \frac{\omega t}{2} \cos \theta \right) |\uparrow\rangle + 2 \sin \frac{\omega t}{2} \sin \theta e^{2i\phi} |\downarrow\rangle$

$$P_{\downarrow} = |\langle \downarrow | \psi(t) \rangle|^2 = \sin^2 \frac{\omega t}{2} \sin^2 \theta$$

$$P_{\uparrow} = |\langle \uparrow | \psi(t) \rangle|^2 = \cos^2 \frac{\omega t}{2} + \sin^2 \frac{\omega t}{2} \cos^2 \theta$$

c) $\theta = \frac{\pi}{4} \Rightarrow P_{\downarrow} = \frac{1}{2} \sin^2 \frac{\omega t}{2}, \quad P_{\uparrow} = \cos^2 \frac{\omega t}{2} + \frac{1}{2} \sin^2 \frac{\omega t}{2}$



$$(P_{\uparrow})_{\max} = 1, (P_{\uparrow})_{\min} = 0.5$$

$$(P_{\downarrow})_{\max} = 0.5, (P_{\downarrow})_{\min} = 0$$

2. Let ψ_{nlm} denote the eigenfunctions of the Hydrogen atom. Suppose the Hydrogen atom is in the state $\psi = \frac{1}{5}[3\psi_{211} + \psi_{310} - \sqrt{15}\psi_{21-1}]$

- a) Find the expectation value of energy as a fraction of ground state energy in the state ψ . [2]
 b) Find the expectation values $\langle L^2 \rangle$ and $\langle L_z \rangle$ in the state ψ . [5]
 c) Find the expectation value $\langle L_x^2 + L_y^2 \rangle$ in the state ψ . [3]

$$a) \frac{\langle \hat{H} \rangle}{E_g} = \frac{\left(\frac{3}{5}\right)^2 \left(\frac{-13.6}{4}\right) + \left(\frac{1}{5}\right)^2 \left(\frac{-13.6}{9}\right) + \frac{15}{25} \left(\frac{-13.6}{4}\right)}{-13.6} = .2445$$

$$b) \left. \begin{aligned} L^2 \psi_{211} &= 2\hbar^2 \psi_{211} \\ L^2 \psi_{310} &= 2\hbar^2 \psi_{310} \\ L^2 \psi_{21-1} &= 2\hbar^2 \psi_{21-1} \end{aligned} \right\} \begin{aligned} L_z \psi_{211} &= \hbar \psi_{211} \\ L_z \psi_{310} &= 0 \\ L_z \psi_{21-1} &= -\hbar \psi_{21-1} \end{aligned}$$

$$\langle L^2 \rangle = \left(\frac{3}{5}\right)^2 \cdot 2\hbar^2 + \left(\frac{1}{5}\right)^2 2\hbar^2 + \frac{15}{25} \cdot 2\hbar^2 = 2\hbar^2$$

$$\langle L_z \rangle = \left(\frac{3}{5}\right)^2 \hbar + \frac{15}{25} (-\hbar) = -\frac{6}{25} \hbar$$

$$c) \left. \begin{aligned} L_z^2 \psi_{211} &= \hbar^2 \psi_{211} \\ L_z^2 \psi_{310} &= 0 \\ L_z^2 \psi_{21-1} &= \hbar^2 \psi_{21-1} \end{aligned} \right\} \langle L_z^2 \rangle = \left(\frac{3}{5}\right)^2 \hbar^2 + \frac{15}{25} \hbar^2 = \frac{24}{25} \hbar^2$$

$$\begin{aligned} \langle L_x^2 + L_y^2 \rangle &= \langle L^2 - L_z^2 \rangle = \langle L^2 \rangle - \langle L_z^2 \rangle \\ &= 2\hbar^2 - \frac{24}{25} \hbar^2 = \frac{26}{25} \hbar^2 \end{aligned}$$